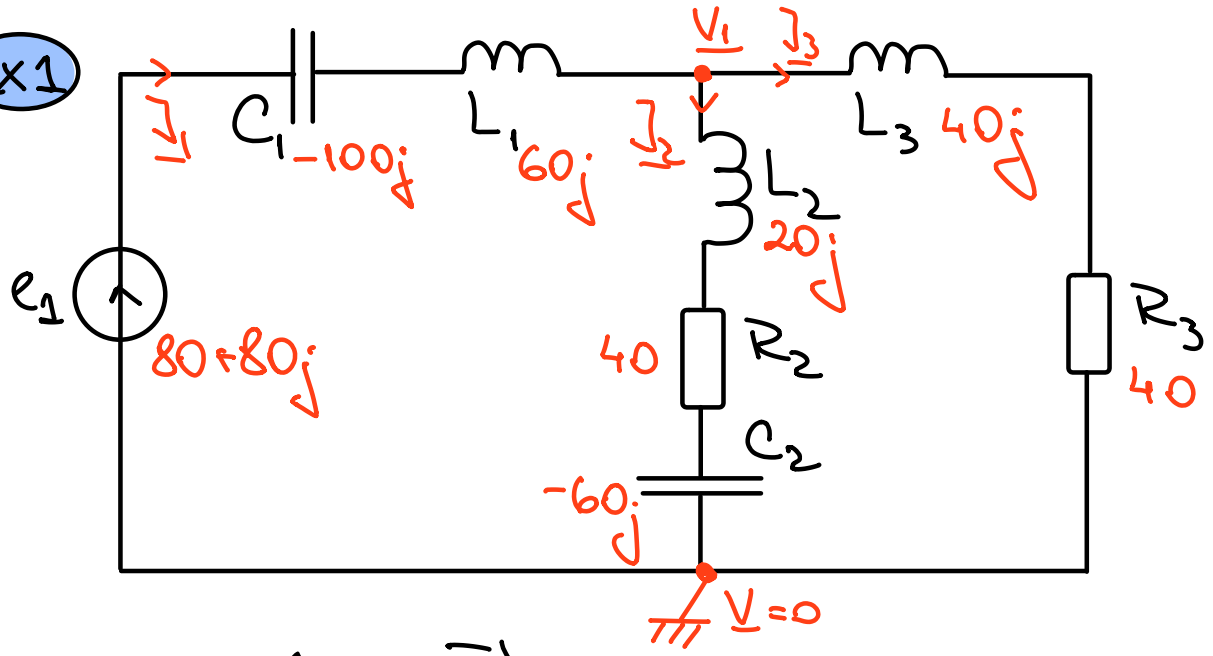
ex1



$e_1 = 160 \operatorname{sen}\left(100t + \frac{\pi}{4}\right)\ V$

$C_1 = 0{,}1\ mF$ $\quad L_1 = 0{,}6\ H$ $\quad L_3 = 0{,}4\ H$

$C_2 = \frac{1}{6}\ mF$ $\quad L_2 = 0{,}2\ H$ $\quad R_2 = R_3 = 40\ \Omega$

$$\underline{Z}_{L_1} = j\omega L_1 = j \cdot 100 \cdot 0{,}6 = 60j$$

$$\underline{Z}_{L_2} = j\omega L_2 = j \cdot 100 \cdot 0{,}2 = 20j$$

$$\underline{Z}_{L_3} = j\omega L_3 = j \cdot 100 \cdot 0{,}4 = 40j$$

$$\underline{Z}_{C_1} = -\frac{j}{\omega C_1} = \frac{-j}{100 \cdot 0{,}1 \cdot 10^{-3}} = -100j$$

$$\underline{Z}_{C_2} = -\frac{j}{\omega C_2} = \frac{-j}{100 \cdot \frac{1}{6} \cdot 10^{-3}} = -60j$$

$$\underline{Z}_{R_2} = \underline{Z}_{R_3} = 40$$

$$\underline{E}_1 = \frac{160}{\sqrt{2}} \cdot e^{j\frac{\pi}{4}} = \frac{160}{\sqrt{2}} \cdot \left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) = 80 + 80j$$

$$-\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0 \Rightarrow -\frac{0 - \underline{V}_1 + \underline{E}_1}{\underline{Z}_{C_1} + \underline{Z}_{L_1}} + \frac{\underline{V}_1 - 0}{\underline{Z}_{L_2} + \underline{Z}_{R_2} + \underline{Z}_{C_2}} + \frac{\underline{V}_1 - 0}{\underline{Z}_{R_3} + \underline{Z}_{L_3}} = 0$$

$$\frac{\underline{V}_1 - \underline{E}_1}{\underline{Z}_{C_1} + \underline{Z}_{L_1}} + \frac{\underline{V}_1}{\underline{Z}_{L_2} + \underline{Z}_{R_2} + \underline{Z}_{C_2}} + \frac{\underline{V}_1}{\underline{Z}_{R_3} + \underline{Z}_{L_3}} = 0$$

$$\Rightarrow \frac{\underline{V}_1 - 80 - 80j}{-100j + 60j} + \frac{\underline{V}_1}{20j + 40 - 60j} + \frac{\underline{V}_1}{40j + 40} = 0$$

$$\Rightarrow \frac{\underline{V}_1 - 80 - 80j}{40j} + \frac{\underline{V}_1}{40 - 40j} + \frac{\underline{V}_1}{40j + 40} = 0 \;/ \cdot 40$$

$$\frac{\underline{V}_1 - 80 - 80j}{j} + \frac{\underline{V}_1}{1 - j} + \frac{\underline{V}_1}{1 + j} = 0$$

$$j\underline{V}_1 - 80j + 80 + \frac{1}{2}\underline{V}_1(1+j) + \frac{1}{2}\underline{V}_1(1-j) = 0$$

$$j\underline{V}_1 - 80j + 80 + \frac{1}{2}\underline{V}_1 \cdot 2 = \underline{V}_1 + j\underline{V}_1 = -80 + 80j \Rightarrow$$

$$\Rightarrow \underline{V}_1(1+j) = 80(-1+j) \Rightarrow \underline{V}_1 = 80 \cdot \frac{-1+j}{1+j}$$

$$\underline{V}_1 = 80 \cdot \frac{-1+j+j-(-1)}{2j} = 80 \cdot j$$

$$\underline{I}_1 = \frac{-80j + 80 + 80j}{60j - 100j} = \frac{80}{-40j} = -\frac{2}{j} = 2j$$

$$i_1(t) = 2\sqrt{2}\sin(100t + \pi/2)$$

$$\underline{I}_2 = \frac{80j}{40 + 20j - 60j} = \frac{80j}{40(1-j)} = \frac{2j(1+j)}{2} = -1 + j$$

$$i_2(t) = 2\sqrt{2}\sin\left(100t + \frac{3\pi}{4}\right)$$

$$\underline{I}_3 = \frac{80j}{40 + 40j} = \frac{2j}{1+j} = \frac{2j(1-j)}{2} = 1 + j$$

$$i_3(t) = 2\sqrt{2}\sin\left(100t + \frac{\pi}{4}\right)$$

Puterea activă $P = UI\cos\varphi$ [W]

Puterea reactivă $Q = UI\sin\varphi$ [VAR]

Puterea aparentă $S = UI$ [VA]

Puterea aparentă complexă $\underline{S} = \underline{U}\cdot\underline{I}^* = P + jQ$

$$\underline{S}_R = \underline{U}_R\cdot\underline{I}_R^* = R\cdot\underline{I}_R\cdot\underline{I}_R^* = R\cdot|\underline{I}_R|^2$$
$$\underline{S}_L = \underline{U}_L\cdot\underline{I}_L^* = j\omega L\,\underline{I}_L\cdot\underline{I}_L^* = j\omega L|\underline{I}_L|^2$$
$$\underline{S}_C = \underline{U}_C\cdot\underline{I}_C^* = \frac{-j}{\omega C}|\underline{I}|^2$$

Bilanțul puterilor $S_{gen} = S_{cons}$

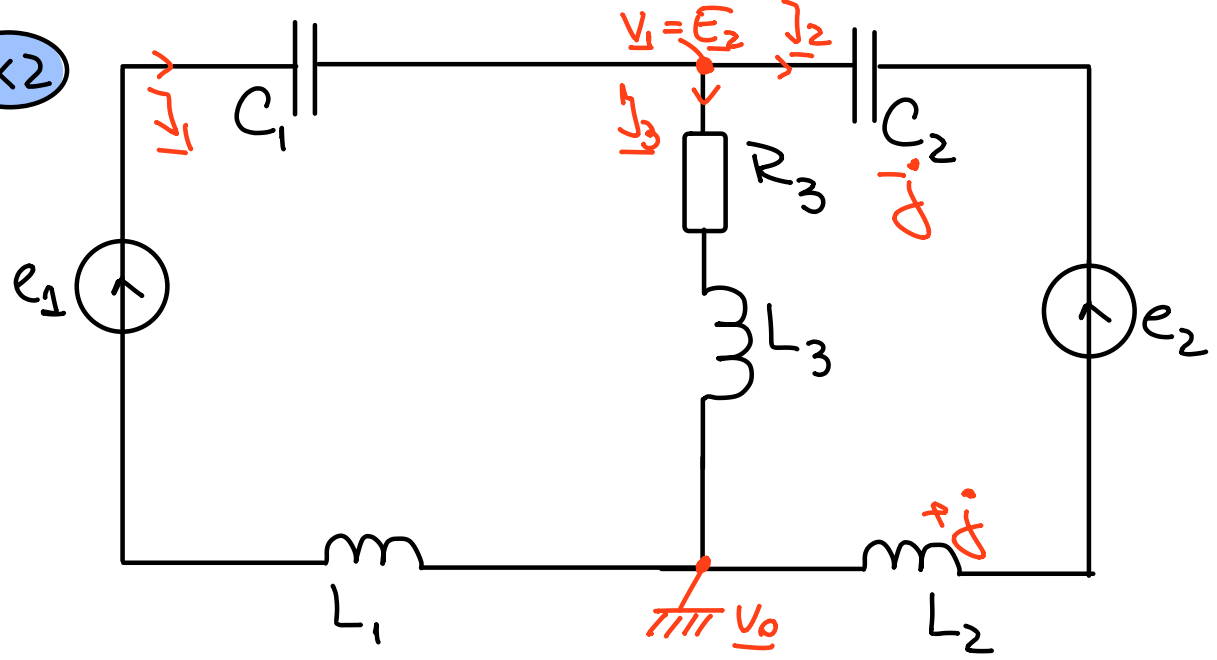
$$\underline{S}_{gen} = \underline{E}_1\cdot\underline{I}_1^* = (80+80j)(-2j) = 160 - 160j$$

$$P_{cons} = R_2 I_2^2 + R_3 I_3^2 = 40(\sqrt{2})^2 + 40(\sqrt{2})^2 = 160\,W$$

$$Q_{cons} = \omega L_1 I_1^2 + \omega L_2 I_2^2 + \omega L_3 I_3^2 - \frac{1}{\omega C_1} I_1^2 - \frac{1}{\omega C_2} I_2^2$$
$$= 60\cdot 4 + 20\cdot(\sqrt{2})^2 + 40(\sqrt{2})^2 = 100\cdot 4 - 60(\sqrt{2})^2$$
$$= -160\ var$$

$$\underline{S}_{cons} = 160 - 160j$$

ex2



$e_1 = 10\cos\left(\omega t - \frac{\pi}{4}\right) V = 10\sin\left(\omega t + \frac{\pi}{4}\right)$

$e_2 = 10\sin\left(\omega t + \frac{3\pi}{4}\right) V$

$\omega L_1 = 3\Omega \qquad \omega L_2 = \omega L_3 = 1\Omega$

$\frac{1}{\omega C_1} = \frac{1}{\omega C_2} = 1\Omega \qquad R_3 = 1\Omega$

$\underline{Z}_{L_1} = j\omega L_1 = 3j \qquad \underline{Z}_{C_1} = -\frac{j}{\omega C_1} = -j$

$\underline{Z}_{L_2} = j\omega L_2 = j \qquad \underline{Z}_{C_2} = -\frac{j}{\omega C_2} = -j$

$\underline{Z}_{L_3} = j\omega L_3 = j \qquad \underline{Z}_{R_3} = 1$

$\underline{E}_1 = \frac{10}{\sqrt{2}} e^{-j\frac{\pi}{4}} = \frac{10}{\sqrt{2}}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) = -5 + 5j$

$\underline{E}_2 = \frac{10}{\sqrt{2}} e^{j\frac{\pi}{4}} = \frac{10}{\sqrt{2}}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) = 5 + 5j$

Obs: E_2 reversă ideală $\Rightarrow \underline{V}_1 = \underline{E}_2 = -5 + 5j$

$\underline{Y}_1 = \frac{0 - \underline{V}_1 + \underline{E}_1}{\underline{Z}_{L_1} + \underline{Z}_{C_1}} = \frac{5 + 5j + 5 - 5j}{3j - j} = \frac{10}{2j} = -5j$

$$i_1(t) = \sqrt{2}\cdot 5 \sin\left(\omega t - \frac{\pi}{2}\right)$$

$$\underline{I}_3 = \frac{\underline{V}_1}{\underline{Z}_{R_3} + \underline{Z}_{L_3}} = \frac{\cancel{1-j}\,-5+5j}{1+j} = \frac{-5+5j+5j+5}{2} = 5j$$

$$i_3(t) = \sqrt{2}\cdot 5 \cdot \sin\left(\omega t + \frac{\pi}{2}\right)$$

$$\underline{I}_2 = \underline{I}_3 - \underline{I}_1 = 5j + 5j = 10j$$

$$i_2(t) = \sqrt{2}\cdot 10 \cdot \sin\left(\omega t + \frac{\pi}{2}\right)$$

Bilanț: $\underline{S}_{gen} = \underline{E}_1 \cdot \underline{I}_1^* + \underline{E}_2 \underline{I}_2^* = (5+5j)\cdot 5j + (-5+5j)(-10j)$

$$= -25 + 25j + 50 + 50j = 25 + 75j$$

$$\Rightarrow P_{gen} = 25\,W$$

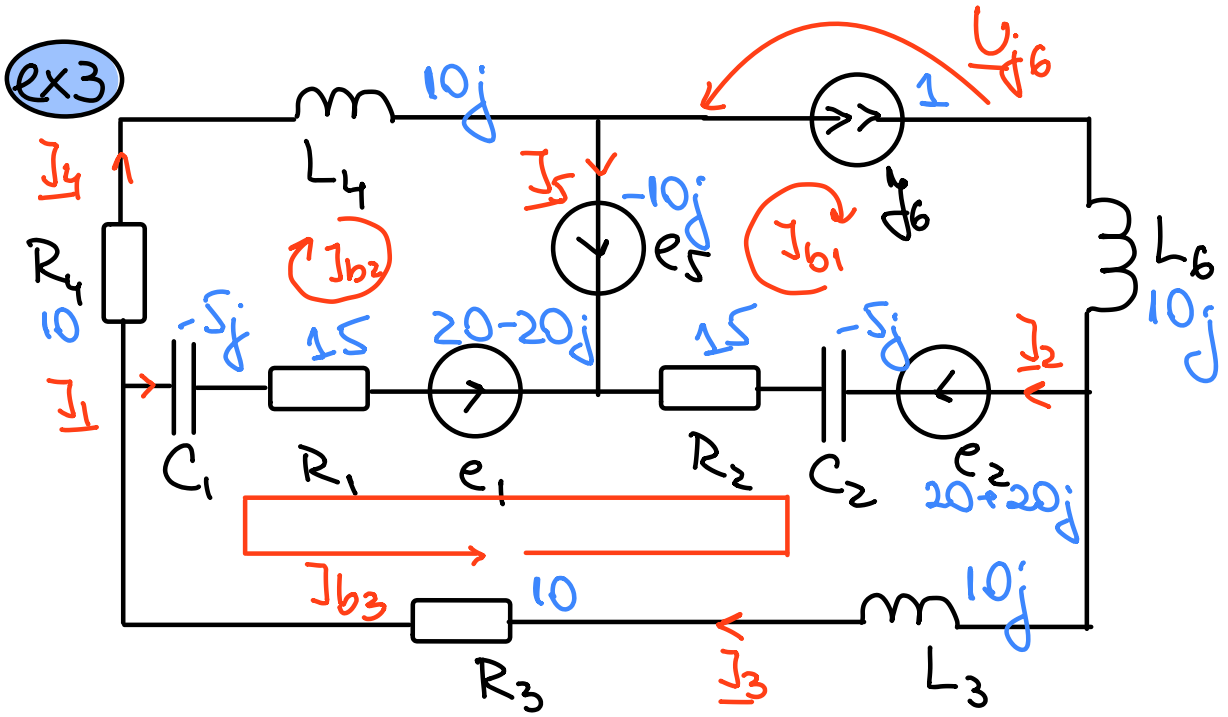
$$Q_{gen} = 75\,VAR$$

$$\underline{S}_{cons} = P_{cons} + jQ_{cons}$$

$$P_{cons} = R_3 I_3^2 = 1\cdot 25 = 25\,W$$

$$Q_{cons} = \omega L_1 I_1^2 + \omega L_2 I_2^2 + \omega L_3 I_3^2 - \frac{1}{\omega C_1} I_1^2 - \frac{1}{\omega C_2} I_2^2$$

$$= 3\cdot 25 + 1\cdot 100 + 1\cdot 25 - 1\cdot 25 - 1\cdot 100 = 75\,VAR$$



$R_1 = 15\,\Omega$ $\quad\quad$ $\omega L_3 = \omega L_4 = \omega L_6 = 10\,\Omega$

$R_2 = 15\,\Omega$ $\quad\quad$ $\frac{1}{\omega C_1} = \frac{1}{\omega C_2} = 5\,\Omega$

$R_3 = 10\,\Omega$

$R_4 = 10\,\Omega$

$e_1 = 40 \sin\left(\omega t - \frac{\pi}{4}\right) V$

$e_2 = 40 \sin\left(\omega t + \frac{\pi}{4}\right) V$

$e_5 = 10\sqrt{2} \sin\left(\omega t - \frac{\pi}{2}\right) V$

$J_6 = \sqrt{2} \sin \omega t \; A$

$\underline{Z}_{R_1} = \underline{Z}_{R_2} = 15$

$\underline{Z}_{R_3} = \underline{Z}_{R_4} = 10$

$\underline{Z}_{L_3} = \underline{Z}_{L_4} = \underline{Z}_{L_6} = 10j$

$\underline{Z}_{LC_1} = \underline{Z}_{LC_2} = -5j$

$\underline{E}_1 = \frac{40}{\sqrt{2}}\left(\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}\right) = 20 - 20j$

$\underline{E}_2 = \frac{40}{\sqrt{2}}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) = 20 + 20j$

$\underline{E}_5 = \frac{10\sqrt{2}}{\sqrt{2}}(0 - j) = -10j$

$\underline{J}_6 = \frac{\sqrt{2}}{\sqrt{2}}(1 + 0) = 1$

$$\begin{cases} \underline{V}_{b1} = \underline{V}_6 = 1 \\ \underline{V}_{b2}(10+10j) + (\underline{V}_{b2} + \underline{V}_{b3})(15-5j) + 10j + 20 - 20j = 0 \\ \underline{V}_{b3}(10+10j) + (\underline{V}_{b3} + \underline{V}_{b2})(15-5j) + 20 - 20j + (\underline{V}_{b3} + \underline{V}_{b1})(15 \\ \quad -5j) - 20 - 20j = 0 \end{cases}$$

$$\begin{cases} \underline{V}_{b1} = \underline{V}_6 = 1 \\ \underline{V}_{b2}(25+5j) + \underline{V}_{b3}(15-5j) = -20+10j \\ \underline{V}_{b2}(15-5j) + \underline{V}_{b3} \cdot 40 = 45j - 15 \end{cases}$$

$$\begin{cases} \underline{V}_{b2}(5+j) + \underline{V}_{b3}(3-j) = -4+2j \\ \underline{V}_{b2}(3-j) + \underline{V}_{b3} \cdot 8 = 9j - 3 \end{cases} \longrightarrow \underline{V}_{b3} = \frac{-3+9j - \underline{V}_{b2}(3-j)}{8}$$

$$\underline{V}_{b2}(5+j) + \frac{-3+9j - \underline{V}_{b2}(3-j)}{8}(3-j) = -4+2j$$

$$\underline{V}_{b2}(40+8j) + (-3+9j)(3-j) - (3-j)^2 \underline{V}_{b2} = (-4+2j)8$$

$$\underline{V}_{b2}(40+8j) - (9-6j-1)\underline{V}_{b2} + (-9+3j+27j+9) = -32+16j$$

$$\underline{V}_{b2}(32+14j) = -32+16j-30j \Rightarrow \underline{V}_{b2} = \frac{-32-14j}{32+14j} = -1$$

$$\underline{V}_{b3} = \frac{-3+9j-(-1)(3-j)}{8} = \frac{-3+9j+3-j}{8} = j$$

$$\underline{I}_1 = -\underline{I}_{b2} - \underline{I}_{b3} = 1 - j \Rightarrow i_1(t) = 2\sin\left(\omega t - \frac{\pi}{4}\right)$$

$$\underline{I}_2 = \underline{I}_{b3} + \underline{I}_{b1} = 1 + j \Rightarrow i_2(t) = 2\sin\left(\omega t + \frac{\pi}{4}\right)$$

$$\underline{I}_3 = -\underline{I}_{b3} = -j \Rightarrow i_3(t) = \sqrt{2}\sin\left(\omega t - \frac{\pi}{2}\right)$$

$$\underline{I}_4 = \underline{I}_{b2} = -1 \Rightarrow i_4(t) = \sqrt{2}\sin(\omega t + \pi)$$

$$\underline{I}_5 = \underline{I}_{b2} - \underline{I}_{b1} = -1 - 1 = -2 \Rightarrow i_5(t) = \sqrt{2}\cdot 2\sin(\omega t + \pi)$$

$$\underline{U}_{j6} - \underline{E}_5 - \underline{I}_2(-5j + 15) + \underline{E}_2 - \underline{I}_6 \cdot 10j = 0$$

$$\underline{U}_{j6} = -10j + (1+j)(-5j+15) - 20 - 20j + 1\cdot 10j$$

$$= -5j + 15 + 5 + 15j - 20 - 20j = -10j$$

$$\underline{S}_{gen} = \underline{E}_1 \cdot \underline{I}_1^*$$

$$\underline{S}_{gen} = \underline{E}_1 \underline{I}_1^* + \underline{E}_2 \underline{I}_2^* + \underline{E}_5 \underline{I}_5^* + \underline{U}_{j6} \cdot \underline{I}_6^*$$

$$= (20 - 20j)(1+j) + (20 + 20j)(1-j) + (-10j)(-2) + (-10j)\cdot 1$$

$$= 20 + 20j - 20j + 20 + 20 - 20j + 20j + 20 + 20j - 10j$$

$$= 80 + 10j$$

$$S_{cons} = (R_1 I_1^2 + R_2 I_2^2 + R_3 I_3^2 + R_4 I_4^2) + j\left(\omega L_3 I_3^2 + \omega L_4 I_4^2 + \omega L_6 I_6^2 - \frac{1}{\omega C_1} I_1^2 - \frac{1}{\omega C_2} I_2^2\right)$$

$$= (15\cdot 2 + 15\cdot 2 + 10\cdot 1 + 10\cdot 1) + j(10\cdot 1 + 10\cdot 1 + 10\cdot 1 - 5\cdot 2 - 5\cdot 2)$$

$$= (30 + 30 + 20) + j(30 - 10 - 10) = 80 + 10j$$